

Blockchain for Supply Chain Transparency and Trust in Enterprise Systems

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Abstract — *This paper presents the integration of blockchain technology into enterprise systems, specifically supply chain to enhance data transparency and trust. It investigates the background, workings and real-world applications of blockchain technology. The study concludes that blockchain has a great potential to support supply chains in enterprise systems.*

Keywords— *Blockchain, Supply Chain, Enterprise System, Transparency, Security*

EXECUTIVE SUMMARY

This paper explores the integration of blockchain technology into enterprise systems to maintain the transparency and security of the information. The first part of the report introduces what blockchain is, continues with investigates its background and discusses challenges while implementing it into current enterprise systems. Blockchain, initially developed to support Bitcoin as a decentralized platform to store all the data securely and keep it synchronized. Until now, its application has then extended to various industries other than the financial field, such as healthcare, supply chain management and logistics. Although blockchain offers many benefits, it is still not widely adopted by most companies, mainly due to the lack of technical expertise and infrastructure support.

The second part of the report explains blockchain's core mechanisms and examines how it improves the transparency and trust for the enterprise performance. Furthermore, the report also reviews the real-world application across various domains, discussing the advantages and disadvantages of blockchain adoption and lastly provides some recommendations for the enterprise. In conclusion, blockchain technology has shown great potential to improve transparency, and trust for supply chains in enterprise systems.

1. INTRODUCTION

In 2009, blockchain first gained popularity with the introduction of Bitcoin in the market and was originally designed to prevent double spending in digital currencies. It began as a secure, transparent, and decentralized platform for peer-to-peer transactions and the recording of information, also known as the foundation of cryptocurrency (Popovski, Soussou, & Webb, 2018). This technology has grown rapidly and become more crucial in enterprise systems, particularly in supply chain management because transparency and trust are important elements for effective collaboration between stakeholders. As an example, tracking goods, verifying transactions are covered by blockchain to ensure accurate and reliable information shared across organizations, especially for multi-partner organizations.

As technology evolves, blockchain becomes a great idea to enhance enterprise operational capabilities. In a recent quantitative investigation, blockchain sharing methods are being proven that perform better than traditional ways in every aspect, including information-sharing efficiency, data accuracy and cost reduction (Jiang, 2024). These improvements can significantly reduce operational errors, allow faster decision making, better resource monitoring and more reliable reporting. Besides to boost the organizational performance and facilitate the business operations, blockchain technology can also lead to transparency and trust improvement within enterprises. In short, blockchain technology acts as a strong foundation for companies systems by keeping their data secure and supporting automated processes.

However, the use of the blockchain in enterprise systems remains limited with several types of issues, such as organizational, ethical, and technical difficulties. In order to better understand blockchain technology, this article will primarily examine its architecture, use cases, benefits, limitations and potential future development. It also aims to offer insight to researchers and organizations who are interested about how blockchain is being used in enterprise settings.

2. BACKGROUND

A blockchain defined as a distributed database shared across a computer nodes network. It is best known in cryptocurrency systems such as Bitcoin, where it enables a safe and decentralized record of transactions (Hayes, 2025). However, blockchain technology is definitely not only limited to cryptocurrency, it also involves finance, healthcare, supply chain management, etc. These characteristics explain why it has gained popularity by producing transparent, tamper-proof, and verifiable transaction records without relying on centralized authorities.

Blockchain technology (BCT) is currently being considered a transformative innovation to increase transparency, traceability, and trust in enterprise systems, particularly supply chain management. Traditional supply chains before this relied on centralized databases, which are often vulnerable to fraud, forgery, and delays (Wu et al., 2019) because the stakeholders like suppliers, manufacturers, distributors, and retailers have restricted exposure into accurate, secure information. However, the decentralized architecture of BCT that securely and consistently records every transaction ensures the same truth for them. (Hariyani et al., 2025).

The studies show that BCT may solve inefficiencies in traditional supply chain systems. For example, in Francisco and Swanson (2018), the researchers study blockchain adoption and evaluate user acceptance using the Unified Theory of Acceptance and Use of Technology (UTAUT) framework based on performance expectancy, effort expectancy, social influence, and facilitating conditions. According to the findings, many organizations are still unwilling to implement BCT due to uncertain return on investment, unclear regulatory from administrative and most important, the lack of technical expertise to integrate it with legacy systems.

Additionally, the researchers also examined how BCT interacts with other technologies to enhance enterprise performance. In the era of Industry 4.0, BCT combines with artificial intelligence (AI) and the Internet of Things (IoT) to provide secure and immutable data for predictive maintenance and real-time inventory management (Hariyani et al., 2025). Besides, smart contracts are used to integrate BCT with automation to support the Circular Economy and sustainable energy trade. As a result, an organization that integrates BCT with other digital technologies becomes more flexible and efficient.

In summary, BCT offers a better solution to solve the problems faced by traditional supply chains. Although its integration with emerging technologies such as AI and IoT enhances enterprise performance and data integrity, the adoption issues still remain. Therefore, future research should focus on developing standardized frameworks and practical integration strategies to fully utilize blockchain.

3. ANALYSIS AND DISCUSSION

3.1 The Blockchain Technology

Blockchain is a trustless, peer-to-peer and ever-expanding ledger of records, including smart contracts, which are executed and shared among involved parties (Hamida et al., 2017). Blockchain, as its name suggests, is a chain of blocks that are cryptographically linked forming a ledger, with each block containing a group of cryptographically signed transactions (ACT-IAC, 2017; Yaga et al., 2018) as shown in Fig. 1.

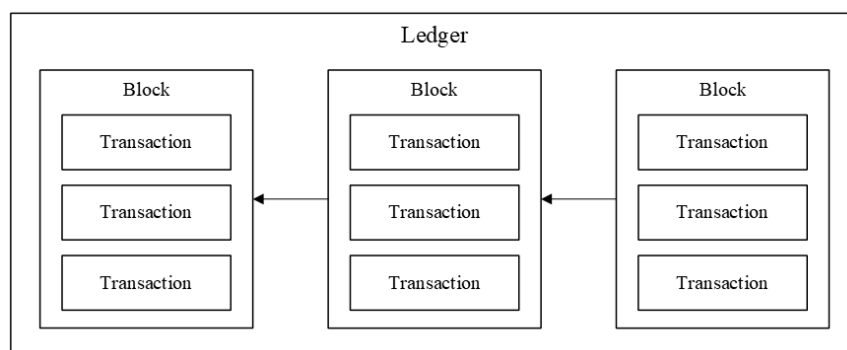


Fig. 1 Transactions are grouped into blocks and referred to as the ledger.

Blockchain relies on peer-to-peer networks. The entire blockchain network is made of peers or nodes. The ledger is distributed to each node in the blockchain network (Yaga et al., 2018) as shown in Fig 2.

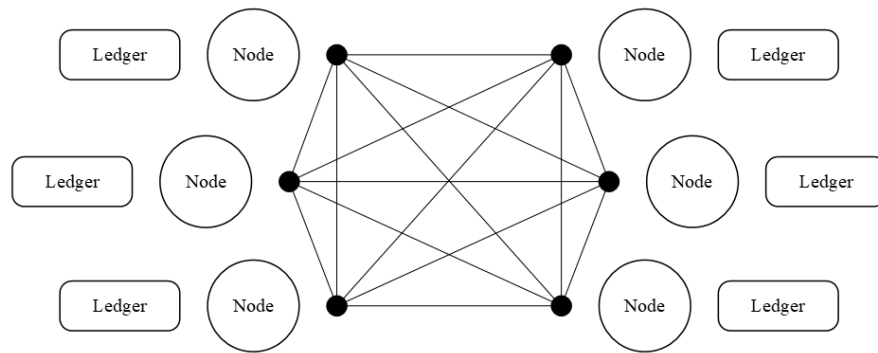


Fig. 2 A blockchain network made up of nodes, with each node holding a copy of the ledger (Murtaza, 2023).

A blockchain network user submits candidate transactions using softwares such as applications, digital wallets and web services. These softwares send the transactions to one or more nodes within the blockchain network. The submitted transactions are then propagated to other nodes across the network. Once it reaches a publishing node, it must wait in a queue until the publishing node adds it to a block and publishes the block to the blockchain. Multiple publishing nodes may compete to publish the next block at the same time and a consensus model such as Proof Of Work and Proof of Stake is used to determine which node publishes the next block. The transaction is validated by ensuring it is correctly formatted and cryptographically signed by the provider of the digital asset. Invalid transactions are ignored, preventing them from propagating throughout the network. Other nodes will check the validity of the transactions in the published block before adding the block to its own copy of the ledger (Yaga et al., 2018).

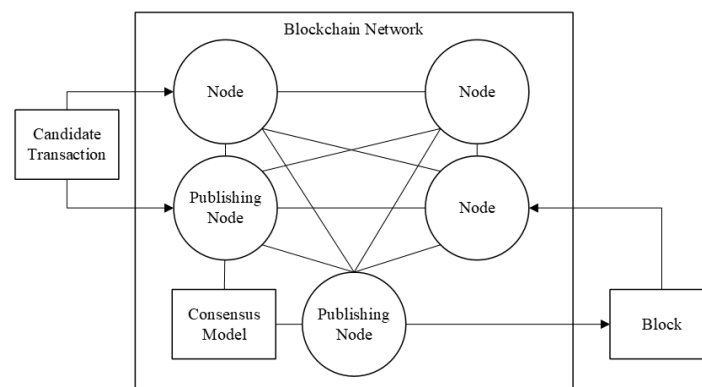


Fig. 3 A transaction is added to a block

3.2 How Blockchain is Transparent and Reliable

The architecture of blockchain is designed for use in environments where participants may not fully trust one another. The security measures and transparent record keeping are the advantages of blockchain technology. The following components help ensure that blockchain is secure and reliable.

1) *Cryptographic hash functions*: Blockchain technology utilises cryptographic hash functions for many operations. Input data is processed to generate a unique output or digest. Cryptographic hash are preimage resistant, where the input cannot be computed given the digest; and collision resistant, where one cannot find two inputs that hash to the same output (Yaga et al., 2018).

2) *Asymmetric-key cryptography*: This method uses a mathematically related pair of public and private keys. Despite the relationship, the private key cannot be efficiently derived from the public key. Each transaction must be 'digitally signed', where the private key encrypts the transaction and anyone with the public key can decrypt it and thus, verifying the signature generated by the private key. Encrypting the transaction with the private key proves that the signer has access to the private key. (Yaga et al., 2018)

3) *Ledgers*: A ledger is a collection of transactions and is distributed among the network, with each user maintaining their own copy. When new nodes join the blockchain network, they discover other full nodes and request a full copy of the ledger. This creates many backup copies and all nodes updating and syncing the same ledger data. Due to its distributed nature, the blockchain network is heterogeneous, where the hardware,

software and network infrastructure vary between nodes. An attack on a node may not work on other nodes. Additionally, there is no centralised point of attack. (Yaga et al., 2018)

4) *Blocks*: A block consists of a block header and block data. The block header stores metadata for the block and the block data contains the list of validated candidate transactions or other data. Blocks are chained together and each block contains the hash digest of the previous block's header. If a previously published block is changed, it will generate a different hash and all subsequent blocks will also have different hashes. This makes altered blocks easily detected. (ACT-IAC, 2017; Yaga et al., 2018) Fig 4 shows a generic chain of blocks.

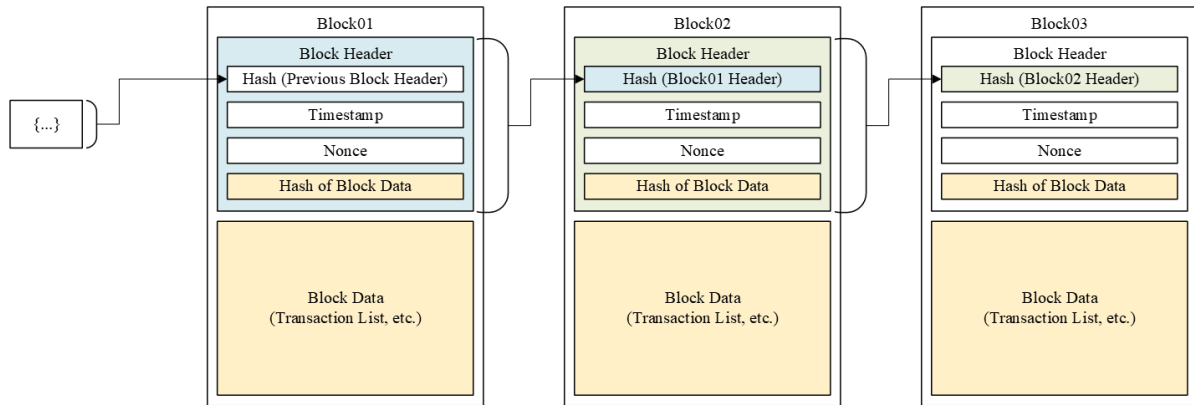


Fig. 4 Generic chain of blocks (Yaga et al., 2018)

The use of cryptographic mechanisms such as cryptographic hash functions and digital signatures ensures tamper resistant and tamper evident ledgers. Each node maintains its own copy of the ledger ensuring each node knows when a transaction has occurred. The distributed ledgers and closely linked blocks make it difficult for attackers to easily compromise the network.

3.3 Applications

Due to the immutability, transparency and security features of blockchain, BCT has been applied in various domains including the financial services, insurance sector, healthcare, voting, real estate and supply chain. (Laroiya et al., 2020)

1) *Financial services*: Barclays Bank implemented the transparency and security features of BCT in their processes (Laroiya et al., 2020).

2) *Insurance sector*: Accenture developed a blockchain system to support key processes in the insurance industry (Laroiya et al., 2020).

3) *Healthcare*: Medicalchain is the first healthcare service provider to use BCT for storing and managing automated health records, to deliver an innovative and unique telemedicine experience. MedRec uses BCT to provide secure access to patients' records (Laroiya et al., 2020).

4) *Voting*: A decentralised ledger to store voting data via BCT could prevent result manipulation by a centralised authority. Some blockchain systems in voting include BitCongress, Romotengrity and AgoraVoting (Laroiya et al., 2020).

5) *Real estate*: In the United States, Arizona and Vermont have already recognised smart contracts as a standard for real estate transactions. The BitProperty application uses the smart contract concept to allow anyone to invest in real estate, while Ubiquity implements blockchain to ensure that property data and information are securely recorded (Laroiya et al., 2020).

6) *Supply chain*: Walmart along with IBM has been working on the Hyperledger Fabric blockchain to track food staples from the suppliers to retail store shelves. Other systems include Blockverify that enforces transparency in supply chains by verifying stolen stock, counterfeit products and deceitful transactions; OriginTrain that allows consumers to trace the origin and production of products; De Beers tracks diamonds from the mine to the purchase; AidCoin that is used in nonprofit sectors; Guts that eliminates ticket fraud and secondary ticket markets by making ticket allocation transparent (Laroiya et al., 2020).

3.4 Misconceptions and Limitations of Blockchain

There is a tendency to overhype and overuse most emerging technologies. The following discusses some limitations and misconceptions of BCT.

1) *Immutability*: Many sources claim that blockchain ledgers are immutable, but blockchain is not entirely immutable and can still be modified. It is tamper evident and tamper resistant and therefore trusted. However, it can still be vulnerable to attacks, such as the 51% attack. In this attack, an attacker needs to control more than 51% of the network's resources to surpass the block creation rate of the remaining network participants. While it is possible to carry out this attack on a blockchain network, it is expensive.

2) *Oracle problem*: Blockchain networks work well with data within their own digital systems. (Yaga et al., 2018) However, external data such as stock prices and weather cannot be incorporated with transaction data, as it would be considered untrusted. To provide external data, blockchain networks rely on a trusted third party known as an oracle. An oracle can be any entity capable of providing reliable external data to the blockchain, such as software, sensors, or humans (ACT-IAC, 2017; Caldarelli, 2020). However, there is no way to verify whether the input data accurately reflects real-world events (Yaga et al., 2018).

3) *No trust*: Although blockchain networks are designed to function without requiring trust between users or intermediaries, a significant amount of trust is still necessary. Users must trust the cryptographic technologies utilised, trust that smart contracts operate correctly and without bugs, trust that the users of the network are not secretly colluding and trust that nodes are processing transactions fairly (Yaga et al., 2018)..

4) *Resource usage*: Blockchain networks that utilise the Proof of Work consensus model require significant resource and thus power consuming. (Yaga et al., 2018; Laroiya et al., 2020) Additionally, when a new full node is created, it must download a large portion or all of the blockchain data, which consumes a lot of network bandwidth.

3.5 The Enterprise System

An enterprise system is a vital support system that contains an organisation's data. It can include various modules such as supply chain management, manufacturing, human resource management, accounting and finance, sales, marketing and customer relationship management (Haddara et al., 2021).

Enterprise systems are centralised, while BCT is decentralised. Despite the conflict, the decentralised, immutable, irreversible and near-real-time capabilities are valuable in enterprise system processes. Therefore, blockchain can be used to enhance and reinforce enterprise systems processes. (Haddara et al., 2021)

3.6 The Supply Chain

Supply chain is the process of goods formations, from the raw materials to intermediate goods, and from final products to customers. It is closely tied to supply chain management, which involves managing materials, money, people, and information within and across the supply chain (Shukla et al., 2011). The main goal of introducing blockchain into the system is to prevent counterfeiting or theft.

In the context of blockchain, the goods and related documentation are the items being transacted. A digital "token" is created for physical items upon their creation. The final recipient of the item can then verify the token and trace its origin (Francisco & Swanson, 2018).

4. RECOMMENDATIONS

Blockchain has gained significant attention, largely due to the rise of Bitcoin that uses BCT, and the overemphasis on its immutable and irreversible properties. Many projects and organisations try to adopt the technology, even when it may not be needed (Yaga et al., 2018).

Several guidelines have been developed by agencies and organisations, including the United States Department of Homeland Security (DHS) Science & Technology Directorate, the American Council for Technology and Industry Advisory Council (ACT-IAC) and numerous white papers and articles titled "Do you need a blockchain" (Yaga et al., 2018; Wust & Gervais, 2018; Peck, 2023).

BCT is suitable for a use case that:

- Requires a shared, consistent data store (Yaga et al., 2018; Peck, 2023)
- Needs more than one entity to contribute data (Yaga et al., 2018; Wust & Gervais, 2018; Peck, 2023)
- Never update or delete data records (Yaga et al., 2018)
- Involves control issues over who runs the data store (Yaga et al., 2018; Peck, 2023)
- Requires a tamper proof log of all writes to the data store (Yaga et al., 2018; Wust & Gervais, 2018)

With this, BCT appears to be an ideal fit for supply chain management. Each stage of the supply chain involves different parties and each party may not always be trusted. A reliable and immutable record keeping system is required. Blockchain can provide an irrefutable record of every transaction with each transaction not being updated and deleted. Each party in the process can keep a copy of the record.

Despite its benefits, there are many challenges in implementing blockchain in the supply chain within enterprise. Many components would need to be integrated with existing enterprise systems. In addition, handling large amounts of data storage is required. Over time, as the number of transactions increases, the number of blocks grows, causing higher data storage demands (Solanke, n.d.; Kunduru, 2023).

CONCLUSION

Blockchain is a decentralised, peer-to-peer system that has a continuously expanding distributed ledger composed of blocks connected together by cryptographically signed transactions. Transactions are created, cryptographically signed and submitted to the blockchain network. These transactions are propagated and verified throughout the network before entering a queue to be published. A publishing node competes to add a block through a consensus model. Once the block is published, it is propagated across the network and nodes append it to their respective ledgers.

The append only ledger and the linked hash between blocks ensures that records are hardly mutable. The use of cryptography technology enhances the security of the blockchain. Thus, blockchain can foster transparency and trust in information systems.

The attention garnered by Bitcoin that uses BCT has contributed to the hype to blockchain, causing more studies and exploration of its applications. More and more enterprises and organisations are considering integrating blockchain into their processes. Blockchain has notable performance in supply chain management to trace items across the supply chain that involve many entities that may not always be trusted.

REFERENCES

- ACT-IAC. (2017). Enabling Blockchain Innovation in the U.S. Federal Government. In *ACT-IAC Emerging Technology COI*. American Council for Technology and Industry Advisory Council (ACT-IAC). https://www.actiac.org/system/files/ACT-IAC%20ENABLING%20BLOCKCHAIN%20INNOVATION_3.pdf
- Caldarelli, G. (2020). *Real-world blockchain applications under the lens of the oracle problem. A systematic literature review* (pp. 1–6). 2020 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD). <https://doi.org/10.1109/ictmod49425.2020.9380598>
- Francisco, K., & Swanson, D. (2018). The supply chain has no clothes: Technology Adoption of blockchain for supply chain transparency. *Logistics*, 2(1), 2. <https://doi.org/10.3390/logistics2010002>
- Haddara, M., Norveel, J., & Langseth, M. (2021). Enterprise systems and blockchain Technology: the dormant Potentials. *Procedia Computer Science*, 181, 562–571. <https://doi.org/10.1016/j.procs.2021.01.203>
- Hamida, E. B., Brousmiche, K. L., Levard, H., & Thea, E. (2017, July 23). *Blockchain for Enterprise: Overview, opportunities and challenges*. <https://hal.science/hal-01591859/>
- Hariyani, D., Hariyani, P., Mishra, S., & Sharma, M. K. (2025). A literature review on transformative impacts of blockchain technology on manufacturing management and industrial engineering practices. *Green Technologies and Sustainability*, 3(3), 100169. <https://doi.org/10.1016/j.grets.2025.100169>
- Hayes, A. (2025, March 24). Blockchain Facts: What Is It, How It Works, and How It Can Be Used. Investopedia. <https://www.investopedia.com/terms/b/blockchain.asp>
- Jiang, L. (2024). The use of blockchain technology in enterprise financial accounting information sharing. *PLOS ONE*, 19(2), e0298210–e0298210. <https://doi.org/10.1371/journal.pone.0298210>
- Wust, K., & Gervais, A. (2018). *Do you Need a Blockchain?* (pp. 45–54). 2018 Crypto Valley Conference on Blockchain Technology (CVCBT). <https://doi.org/10.1109/cvcbt.2018.00011>
- Kunduru, A. R. (2023). Blockchain Technology for ERP Systems: A review. *AMERICAN Journal of Engineering, Mechanics and Architecture*, 01(07), e2993-2637.

<https://khaneprozhe.ir/wp-content/uploads/2024/04/BLOCKCHAIN-TECHNOLOGY-FOR-ERP-SYS-TEMS-A-REVIEW.pdf>

- Laroiya, C., Saxena, D., & Komalavalli, C. (2020). Applications of blockchain technology. In *Elsevier eBooks* (pp. 213–243). <https://doi.org/10.1016/b978-0-12-819816-2.00009-5>
- Murtaza, A. (2023, January 18). *What is Blockchain Technology and How Does it Work?* Creative Tim's Blog - Fully Coded Design Resources for Web Developers. <https://www.creative-tim.com/blog/educational-tech/what-is-blockchain-technology-and-how-does-it-work/>
- Peck, M. E. (2023, September 20). Do you need a blockchain? *IEEE Spectrum*. <https://spectrum.ieee.org/do-you-need-a-blockchain>
- Shukla, R. K., Garg, D., & Agarwal, A. (2011). Understanding of Supply Chain: A Literature Review. *International Journal of Engineering Science and Technology (IJEST)*, 3(3), 2059–2072. https://www.researchgate.net/publication/50946316_UNDERSTANDING_OF_SUPPLY_CHAIN_A_LITERATURE_REVIEW
- Solanke, A. (n.d.). *Blockchain implementation for enterprise data integrity: distributed ledger strategies for supply chain transparency and trustworthy data verification* [Dallas Baptist University]. https://www.researchgate.net/profile/Adedamola-Solanke/publication/390212745_Blockchain_Implementation_for_Enterprise_Data_Integrity_Distributed_Ledger_Strategies_for_Supply_Chain_Transparency_and_Trustworthy_Data_Verification/links/67e49df58a5ab03f971310b5/Blockchain-Implementation-for-Enterprise-Data-Integrity-Distributed-Ledger-Strategies-for-Supply-Chain-Transparency-and-Trustworthy-Data-Verification.pdf
- Popovski, L., Soussou, G., & Webb, P. B. (2018). A brief history of blockchain. *Legaltech news*, 3.
- Wu, H., Cao, J., Yang, Y., Tung, C. L., Jiang, S., Tang, B., Liu, Y., Wang, X., & Deng, Y. (2019). Data Management in Supply Chain Using Blockchain: Challenges and a Case Study. 2019 28th International Conference on Computer Communication and Networks (ICCCN). <https://doi.org/10.1109/icccn.2019.8846964>
- Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2018). Blockchain technology overview. In *National Institute of Standards and Technology Internal Report 8202*. <https://doi.org/10.6028/nist.ir.8202>

LOGBOOK

Group 2

Backlog

Team capacity

Current iteration

Roadmap

My items

New view

Add status update

Insi

Assignees

No Assignees

11

Filter by keyword or by field

Title	Status	Intensity	Estimate	Iteration
No Priority 11 Estimate: 0				
1 Do research on our topic - Blockchain for Supply Chain Transparency and Trust in Enterpris... #13	Done	16		Iteration 1
2 Plan the content #5	Done	1		Iteration 1
3 Executive Summary #14	Done	2		Iteration 2
4 Introduction #15	Done	2		Iteration 2
5 Background / Literature Review #16	Done	8		Iteration 2
6 Analysis and Discussion #17	Done	8		Iteration 2
7 Recommendations #18	Done	4		Iteration 2
8 Conclusion #19	Done	2		Iteration 2
9 Tidy up everything #20	In Progress	1		Iteration 3
10 Turnitin Check #98	In Progress	1		Iteration 3
11 Submit Report #21	Todo	1		Iteration 3

Completed

Iteration 1

3 days Oct 19 - Oct 21

Planned

Iteration 2

5 days Oct 22 - Oct 26

Current

Iteration 3

2 days Oct 27 - Oct 28

Do research on our topic - Blockchain for Supply Chain Transparency and Trust in Enterprise Systems

jygoe opened last week

...

Name: Goe

Date & Time: 18 Oct 2025 10.00pm

Keywords Used: Blockchain, Supply Chain Transparency, Technology

Reference: Francisco, K., & Swanson, D. (2018). The Supply Chain Has No Clothes: Technology Adoption of Blockchain for Supply Chain Transparency. Logistics, 2(1), 2. <https://doi.org/10.3390/logistics2010002>

Summary: The article discusses how blockchain enhances transparency and trust in supply chains by creating secure, traceable records of transactions. It highlights real-world uses like tracking tuna and diamonds, and explains factors affecting enterprise adoption of blockchain technology.



Yuylam last week · edited by Yuylam

Edits ▾ ...

Name: Lam

Date & Time: 19 Oct 2025 1.35pm

Platform: Google Scholar

Keywords: Blockchain for Supply Chain Transparency and Trust in Enterprise Systems

Reference: Centobelli, P., Cerchione, R., Del Vecchio, P., Oropallo, E., & Secundo, G. (2021). Blockchain technology for bridging trust, traceability and transparency in circular supply chain. Information & Management, 59(7), 103508.

<https://doi.org/10.1016/j.im.2021.103508>

Platform: YouTube

Keywords: How blockchain works, blockchain use cases

References:

- Simplilearn. (2019, February 27). Blockchain in 7 minutes | What is Blockchain | Blockchain Explained|How Blockchain Works|Simplilearn [Video]. YouTube. <https://www.youtube.com/watch?v=yubzJw0uiE4>
- AI Uncovered. (2023, August 6). Blockchain technology simply explained [Video]. YouTube. <https://www.youtube.com/watch?v=QJn28fFKUR0>
- Simply Explained. (2018, May 29). Blockchains: how can they be used? (Use cases for Blockchains) [Video]. YouTube. https://www.youtube.com/watch?v=aQWfINQuP_o
- Anders Brownworth. (2016, November 5). Blockchain 101 - A visual demo [Video]. YouTube. https://www.youtube.com/watch?v=_160oMzblY8
- Anders Brownworth. (2017, December 30). Blockchain 101 - Part 2 - Public / Private Keys and Signing [Video]. YouTube. https://www.youtube.com/watch?v=xLDL_akeras

Purpose: To gain a better understanding and a clearer picture of how blockchain actually works

Some thought: This reminds me of the KADA system that we did on the previous semester. There are certainly several "grey area" where many transactions are happening, but our system functions more like a data entry tool, recording all the data and transaction information. However, we don't know for certain if the transaction was actually completed or if the recipient received it. I think this is where blockchain could come into play.



jygoe last week

Author ...

Name: Goe

Date & Time: 19 Oct 2025 5.25pm

Tool: ChatGPT

Prompt given: Provide me some article about Blockchain for Supply Chain Transparency and Trust in Enterprise Systems to support my technical report writing.

Purpose of Use:

To search for relevant academic articles and industry papers supporting the topic "Blockchain for Supply Chain Transparency and Trust in Enterprise Systems." Used for idea generation and literature sourcing, not for final content writing.

AI Output Summary:

ChatGPT provided a list of peer-reviewed articles and reviews from reliable publishers (Springer, MDPI, arXiv) discussing blockchain's role in transparency, trust, and enterprise integration within supply chains.

Verification Steps:

1. Cross-checked article links and publication details (title, author, year, journal) on publisher websites.
2. Confirmed each source's relevance to blockchain transparency and enterprise trust before inclusion in the report.





Yuylam last week

...

Name: Lam

Date & Time: 20 Oct 2025 4.35pm

Platform: Google Scholar

Keywords: blockchain in enterprise systems

Reference:

M. Haddara, J. Norveel, and M. Langseth, "Enterprise Systems and Blockchain Technology: The Dormant Potentials," *Procedia Computer Science*, vol. 181, pp. 562–571, Jan. 2021, doi: 10.1016/j.procs.2021.01.203.

Key takeaway: two blockchain characteristics stand out as major factors that can reinforce enterprise systems processes; accomplishing a single source of truth, and a common environment for shared information amongst a larger scope of actors.

Comment: The paper analyses the potential of blockchain through literature review and interview. It discussed the strength of blockchain, which became the reason of why ERP systems would adopt blockchain.

E. B. Hamida, K. L. Brousmiche, H. Levard, and E. Thea, "Blockchain for Enterprise: Overview, opportunities and challenges," Jul. 23, 2017. <https://hal.science/hal-01591859v1>

Personal thoughts: Since blockchain distributes its data to all the actors involved, where is the data stored, and as time passes, the amount of blocks would be increasing. Storage is my current concern. Another concern is the computational cost, as blockchain is more or less like a linked list and searching for a piece of information, means needing to traverse the list. Another part that needs the computational power is where the block is mined.

Comment: The paper discussed the working of blockchain more detailedly compared to the YouTube video saw yesterday. The paper also discussed the scalability of blockchain, which is similar to the question raised.



jygoe last week · edited by jygoe

Edits ▾

Author

...

Name: Goe

Date & Time: 20 Oct 2025 9.20pm

Keywords Used: Blockchain in enterprise systems

Tools: Google Scholar

Reference: Elyes Ben Hamida, Kei Leo Brousmiche, Hugo Levard, Eric Thea. Blockchain for Enterprise: Overview, Opportunities and Challenges. The Thirteenth International Conference on Wireless and Mobile Communications (ICWMC 2017), Jul 2017, Nice, France. fhal-01591859f

Summary: The article provides an in-depth analysis of blockchain technology, focusing on its core components, architecture, consensus mechanisms, and enterprise applications. It explains how blockchain enhances transparency, security, and automation across sectors like finance, energy, logistics, and mobility. However, it also highlights major challenges such as governance, scalability, data privacy, auditability, and security that hinder large-scale enterprise adoption. The paper concludes that blockchain represents a major paradigm shift in enterprise systems, but addressing these challenges is crucial for achieving reliable, scalable, and trustworthy business solutions.



Plan the content



Yuylam last week · edited by Yuylam

Edits ▾

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Date: 21 Oct 2025

Goe

- Executive Summary
- Introduction
- Background / Literature Review
Lam
- Analysis and Discussion
- Recommendations
- Conclusion



Executive Summary



jygoe now

...

Name: Goe

Date & Time: 26 Oct 2025 11.00pm

Read and double check the content of report. Then, complete the executive summary.



Introduction



jygoe 3 days ago

...

Name: Goe

Date & Time: 24 Oct 2025 9.21pm

Tool: ChatGPT

Prompt given: To make my introduction meaningful, aside from the definition of blockchain and the structure of my report, what else of content I can add.

Purpose of use: Find some ideas for introduction to make it more complete

Summary of AI output:

1. Describe some history of blockchain
2. Related blockchain with the topic - supply chain in enterprise system



jygoe 2 days ago

...

Name: Goe

Date & Time: 24 Oct 2025 10.43pm

Keywords Used: The history of blockchain

Tools: Google Scholar

Reference: Popovski, L., Soussou, G., & Webb, P. B. (2018). A brief history of blockchain. Legaltech news, 3.

Summary: The article explain the history of blockchain that starting with the introduction of bitcoin in 2009, until now. After reading the whole article, I concluded some key issues and make it as the first paragraph of my introduction.



jygoe 2 days ago

...

Name: Goe

Date & Time: 25 Oct 2025 00.21pm

Keywords Used: Blockchain in enterprise system

Tools: Google Scholar

Reference: Jiang, L. (2024). The use of blockchain technology in enterprise financial accounting information sharing. PLOS ONE, 19(2), e0298210–e0298210. <https://doi.org/10.1371/journal.pone.0298210>

Summary: The article discuss the use of blockchain in enterprise, clearly describe what is blockchain, how the enterprise use it and also compare the difference between the conventional and blockchain sharing method. As I finished the article, I gain a lot about blockchain and better understand why should we advance our technology rather than stick with outdated methods.



Background



jygoe last week

...

Name: Goe

Date & Time: 21 Oct 2025 9.45pm

Keywords Used: what is blockchain technology

Tools: Google

Reference: Hayes, A. (2025, March 24). Blockchain Facts: What Is It, How It Works, and How It Can Be Used. Investopedia.

<https://www.investopedia.com/terms/b/blockchain.asp>

Comment: I found some definitions and roles of blockchain technology which can be implement into the background of the report



jygoe last week · edited by jygoe

Edits ▾

...

Name: Goe

Date & Time: 22 Oct 2025 00.11am

Keywords Used: Supply Chain Using Blockchain Challenges and Case Study, Blockchain Technology

Tools: Google Scholar

References: Wu, H., Cao, J., Yang, Y., Tung, C. L., Jiang, S., Tang, B., Liu, Y., Wang, X., & Deng, Y. (2019). Data Management in Supply Chain Using Blockchain: Challenges and a Case Study. 2019 28th International Conference on Computer Communication and Networks (ICCCN). <https://doi.org/10.1109/icccn.2019.8846964>

Hariyani, D., Hariyani, P., Mishra, S., & Sharma, M. K. (2025). A literature review on transformative impacts of blockchain technology on manufacturing management and industrial engineering practices. Green Technologies and Sustainability, 3(3), 100169. <https://doi.org/10.1016/j.grets.2025.100169>

Comment: This articles helped me identify the major challenges in traditional supply chain systems and explained how blockchain technology (BCT) can improve data management. They also provide rich insights into the benefits of BCT for transparency.



Analysis and Discussion



Yuylam 5 days ago

Author

...

Name: Lam

Date & Time: 22 Oct 2025

Platform: Google Scholar

Keyword: Blockchain

Reference: D. Yaga, P. Mell, N. Roby, and K. Scarfone, "Blockchain technology overview," Oct. 01, 2018.

<https://doi.org/10.6028/nist.ir.8202>

Key Takeaways:

- Informal Definition of Blockchain: Blockchains are distributed digital ledgers of cryptographically signed transactions that are grouped into blocks. Each block is cryptographically linked to the previous one (making it tamper evident) after validation and undergoing a consensus decision. As new blocks are added, older blocks become more difficult to modify (creating tamper resistance). New blocks are replicated across copies of the ledger within the network, and any conflicts are resolved automatically using established rules.
- Key Characteristics of blockchain: ledger, secure, shared, distributed
- Permissioned and permissionless blockchain [In an enterprise system, the blockchain should be permissioned]
- Components of a blockchain: cryptographic hash functions, transactions, asymmetric-key cryptography, addresses, ledgers, blocks [this may be made into diagram later]
- // Not completed reading the paper





Yuylam 3 days ago

Author ...

Name: Lam

Date & Time: 24 Oct 2025

Completed:

- 3.1 The Blockchain Technology
- 3.2 How Blockchain is Transparent and Reliable



Yuylam 2 days ago

Author ...

Name: Lam

Date & Time: 25 Oct 2025

Platform: Google Scholar

Keyword: blockchain application

Reference: C. Laroia, D. Saxena, and C. Komalavalli, "Applications of Blockchain Technology," in Elsevier eBooks, 2020, pp. 213–243. doi: 10.1016/b978-0-12-819816-2.00009-5.

Platform: Google Scholar

Keyword: oracle problem in blockchain

Reference: G. Caldarelli, Real-world blockchain applications under the lens of the oracle problem. A systematic literature review. 2020 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD), 2020, pp. 1–6. doi: 10.1109/ictmod49425.2020.9380598.

Reference source from "Blockchain technology overview"

K. Wust and A. Gervais, Do you Need a Blockchain? 2018 Crypto Valley Conference on Blockchain Technology (CVCBT), 2018, pp. 45–54. doi: 10.1109/cvcbt.2018.00011.

Platform: Google Scholar

Keyword: Supply chain, blockchain in supply chain

References:

- R. K. Shukla, D. Garg, and A. Agarwal, "Understanding of Supply Chain: A Literature Review," International Journal of Engineering Science and Technology (IJEST), vol. 3, no. 3, Mar. 2011, [Online]. Available: https://www.researchgate.net/publication/50946316_UNDERSTANDING_OF_SUPPLY_CHAIN_A_LITERATURE_REVIEW
- K. Francisco and D. Swanson, "The supply chain has no clothes: Technology Adoption of blockchain for supply chain transparency," Logistics, vol. 2, no. 1, p. 2, Jan. 2018, doi: 10.3390/logistics2010002.

Completed:

3.3 Applications

3.4 Limitations and Misconceptions of Blockchain

3.5 The Supply Chain





Yuylam 17 hours ago

Author ...

Name: Lam

Date & Time: 26 Oct 2025

Platform: Google Scholar

Keyword: enterprise system, blockchain in enterprise system

References:

- Haddara, M., Norveel, J., & Langseth, M. (2021). Enterprise Systems and Blockchain Technology: The Dormant Potentials. Procedia Computer Science, 181, 562–571. <https://doi.org/10.1016/j.procs.2021.01.203>
- Hamida, E. B., Brousmiche, K. L., Levard, H., & Thea, E. (2017, July 23). Blockchain for Enterprise: Overview, opportunities and challenges. <https://hal.science/hal-01591859/>

Completed:

3.6 The Enterprise System



Recommendations



Yuylam 17 hours ago

Author ...

Name: Lam

Date & Time: 26 Oct 2025

Platform: Google Scholar

Keyword: Blockchain for Supply Chain Transparency and Trust in Enterprise Systems

References:

- Solanke, A. (n.d.). Blockchain implementation for enterprise data integrity: distributed ledger strategies for supply chain transparency and trustworthy data verification. https://www.researchgate.net/profile/Adedamola-Solanke/publication/390212745_Blockchain_Implementation_for_Enterprise_Data_Integrity_Distributed_Ledger_Strategies_for_Supply_Chain_Transparency_and_Trustworthy_Data_Verification/links/67e49df58a5ab03f971310b5/Blockchain-Implementation-for-Enterprise-Data-Integrity-Distributed-Ledger-Strategies-for-Supply-Chain-Transparency-and-Trustworthy-Data-Verification.pdf
- Kunduru, A. R. (2023). Blockchain Technology for ERP Systems: A review. AMERICAN Journal of Engineering, Mechanics and Architecture, 1(7). <https://khaneprozhe.ir/wp-content/uploads/2024/04/BLOCKCHAIN-TECHNOLOGY-FOR-ERP-SYSTEMS-A-REVIEW.pdf>

Completed

4. Recommendation



Conclusion



Yuylam 15 hours ago · edited by Yuylam

Edits ▾ Author ...

Date: 26 Oct 2025

Completed Conclusion



Tidy up everything



Yuylam 3 minutes ago

Author ...

Date 27 Oct 2025

Platform: ChatGPT

Used chatgpt to proofread and paraphrase some sentences

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- "My text in the report" paraphrase this

Completed:

Reference rearrangement and organisation



Blockchain for Supply Chain Transparency and Trus...

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Blockchain for Supply Chain Transparency and Trust in Enterprise Systems

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Abstract — This paper presents the integration of blockchain technology into enterprise systems, specifically supply chain to enhance data transparency and trust. It investigates the background, workings and real-world applications of blockchain technology. The study concludes that blockchain has a great potential to support supply chains in enterprise systems.

Keywords— Blockchain, Supply Chain, Enterprise System, Transparency, Security

EXECUTIVE SUMMARY

This paper explores the integration of blockchain technology into enterprise systems to maintain the transparency and security of the information. The first part of the report introduces what blockchain is, continues with investigates its background and discusses challenges while implementing it into current enterprise systems. Blockchain, initially developed to support Bitcoin as a decentralized platform to store all the data securely and keep it synchronized. Until now, its application has then extended to various industries other than the financial field, such as healthcare, supply chain management and logistics. Although blockchain offers many benefits, it is still not widely adopted by most companies, mainly due to the lack of technical expertise and infrastructure support.

The second part of the report explains blockchain's core mechanisms and examines how it improves the transparency and trust for the enterprise performance. Furthermore, the report also reviews the real-world application across various domains, discussing the advantages and disadvantages of blockchain adoption and lastly provides some recommendations for the enterprise. In conclusion, blockchain technology has shown great potential to improve transparency, and trust for supply chains in enterprise systems.

1. INTRODUCTION

In 2009, blockchain first gained popularity with the introduction of Bitcoin in the market and was originally designed to prevent double spending in digital currencies. It began as a secure, transparent, and decentralized platform for peer-to-peer transactions and the recording of information, also known as the foundation of cryptocurrency (Popovski, Soussou, & Webb, 2018). This technology has grown rapidly and become more crucial in enterprise systems, particularly in supply chain management because transparency and trust are important elements for effective collaboration between stakeholders. As an example, tracking goods, verifying transactions are covered by blockchain to ensure accurate and reliable information shared across organizations, especially for multi-partner organizations.

As technology evolves, blockchain becomes a great idea to enhance enterprise operational capabilities. In a recent quantitative investigation, blockchain sharing methods are being proven that perform better than traditional ways in every aspect, including information-sharing efficiency, data accuracy and cost reduction (Jiang, 2024). These improvements can significantly reduce operational errors, allow faster decision making, better resource monitoring and more reliable reporting. Besides to boost the organizational performance and facilitate the business operations, blockchain technology can also lead to transparency and trust improvement within enterprises. In short, blockchain technology acts as a strong foundation for companies systems by keeping their data secure and supporting automated processes.

However, the use of the blockchain in enterprise systems remains limited with several types of issues, such as organizational, ethical, and technical difficulties. In order to better understand blockchain technology, this article will primarily examine its architecture, use cases, benefits, limitations and potential future development. It also aims to offer insight to researchers and organizations who are interested about how blockchain is being used in enterprise settings.

2. BACKGROUND

A blockchain defined as a distributed database shared across a computer nodes network. It is best known in cryptocurrency systems such as Bitcoin, where it enables a safe and decentralized record of transactions (Hayes, 2025). However, blockchain technology is definitely not only limited to cryptocurrency, it also involves finance, healthcare, supply chain management, etc. These characteristics explain why it has gained popularity by producing transparent, tamper-proof, and verifiable transaction records without relying on centralized authorities.

Blockchain technology (BCT) is currently being considered a transformative innovation to increase transparency, traceability, and trust in enterprise systems, particularly supply chain management. Traditional supply chains before this relied on centralized databases, which are often vulnerable to fraud, forgery, and delays (Wu et al., 2019) because the stakeholders like suppliers, manufacturers, distributors, and retailers have restricted exposure into accurate, secure information. However, the decentralized architecture of BCT that securely and consistently records every transaction ensures the same truth for them. (Hariyani et al., 2025).

The studies show that BCT may solve inefficiencies in traditional supply chain systems. For example, in Francisco and Swanson (2018), the researchers study blockchain adoption and evaluate user acceptance using the Unified Theory of Acceptance and Use of Technology (UTAUT) framework based on performance expectancy, effort expectancy, social influence, and facilitating conditions. According to the findings, many organizations are still unwilling to implement BCT due to uncertain return on investment, unclear regulatory from administrative and most important, the lack of technical expertise to integrate it with legacy systems.

Additionally, the researchers also examined how BCT interacts with other technologies to enhance enterprise performance. In the era of Industry 4.0, BCT combines with artificial intelligence (AI) and the Internet of Things (IoT) to provide secure and immutable data for predictive maintenance and real-time inventory management (Hariyani et al., 2025). Besides, smart contracts are used to integrate BCT with automation to support the Circular Economy and sustainable energy trade. As a result, an organization that integrates BCT with other digital technologies becomes more flexible and efficient.

In summary, BCT offers a better solution to solve the problems faced by traditional supply chains. Although its integration with emerging technologies such as AI and IoT enhances enterprise performance and data integrity, the adoption issues still remain. Therefore, future research should focus on developing standardized frameworks and practical integration strategies to fully utilize blockchain.

3. ANALYSIS AND DISCUSSION

3.1 The Blockchain Technology

Blockchain is a trustless, peer-to-peer and ever-expanding ledger of records, including smart contracts, which are executed and shared among involved parties (Hamida et al., 2017). Blockchain, as its name suggests, is a chain of blocks that are cryptographically linked forming a ledger, with each block containing a group of cryptographically signed transactions (ACT-IAC, 2017; Yaga et al., 2018) as shown in Fig. 1.

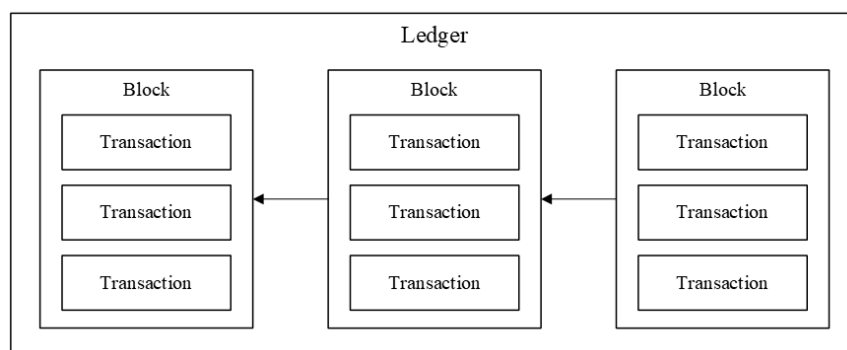


Fig. 1 Transactions are grouped into blocks and referred to as the ledger.

Blockchain relies on peer-to-peer networks. The entire blockchain network is made of peers or nodes. The ledger is distributed to each node in the blockchain network (Yaga et al., 2018) as shown in Fig 2.

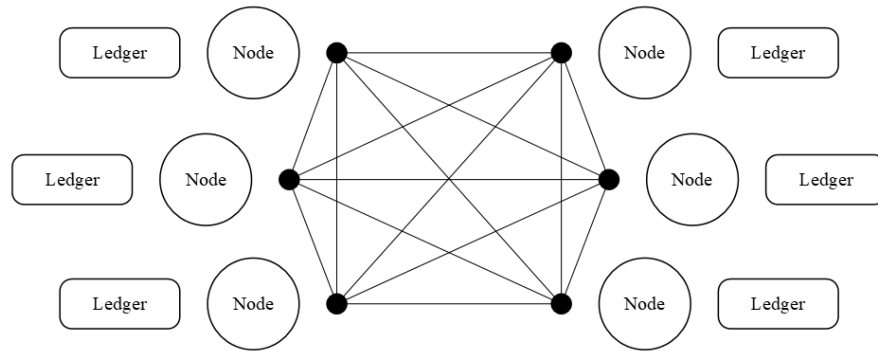


Fig. 2 A blockchain network made up of nodes, with each node holding a copy of the ledger (Murtaza, 2023).

A blockchain network user submits candidate transactions using softwares such as applications, digital wallets and web services. These softwares send the transactions to one or more nodes within the blockchain network. The submitted transactions are then propagated to other nodes across the network. Once it reaches a publishing node, it must wait in a queue until the publishing node adds it to a block and publishes the block to the blockchain. Multiple publishing nodes may compete to publish the next block at the same time and a consensus model such as Proof Of Work and Proof of Stake is used to determine which node publishes the next block. The transaction is validated by ensuring it is correctly formatted and cryptographically signed by the provider of the digital asset. Invalid transactions are ignored, preventing them from propagating throughout the network. Other nodes will check the validity of the transactions in the published block before adding the block to its own copy of the ledger (Yaga et al., 2018).

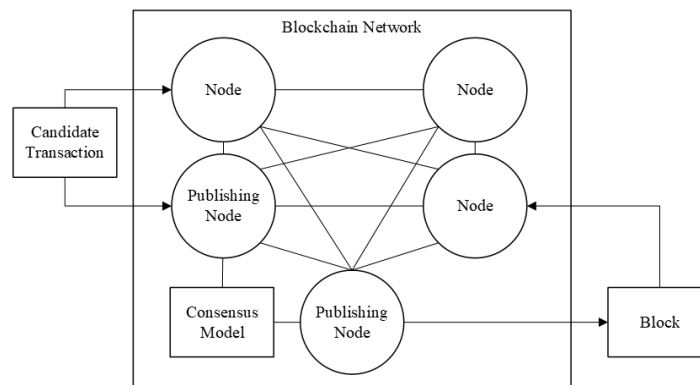


Fig. 3 A transaction is added to a block

3.2 How Blockchain is Transparent and Reliable

The architecture of blockchain is designed for use in environments where participants may not fully trust one another. The security measures and transparent record keeping are the advantages of blockchain technology. The following components help ensure that blockchain is secure and reliable.

1) **Cryptographic hash functions:** Blockchain technology utilises cryptographic hash functions for many operations. Input data is processed to generate a unique output or digest. Cryptographic hash are preimage resistant, where the input cannot be computed given the digest; and collision resistant, where one cannot find two inputs that hash to the same output (Yaga et al., 2018).

2) **Asymmetric-key cryptography:** This method uses a mathematically related pair of public and private keys. Despite the relationship, the private key cannot be efficiently derived from the public key. Each transaction must be 'digitally signed', where the private key encrypts the transaction and anyone with the public key can decrypt it and thus, verifying the signature generated by the private key. Encrypting the transaction with the private key proves that the signer has access to the private key. (Yaga et al., 2018)

3) **Ledgers:** A ledger is a collection of transactions and is distributed among the network, with each user maintaining their own copy. When new nodes join the blockchain network, they discover other full nodes and request a full copy of the ledger. This creates many backup copies and all nodes updating and syncing the same ledger data. Due to its distributed nature, the blockchain network is heterogeneous, where the hardware,

software and network infrastructure vary between nodes. An attack on a node may not work on other nodes. Additionally, there is no centralised point of attack. (Yaga et al., 2018)

4) **Blocks:** A block consists of a block header and block data. The block header stores metadata for the block and the block data contains the list of validated candidate transactions or other data. Blocks are chained together and each block contains the hash digest of the previous block's header. If a previously published block is changed, it will generate a different hash and all subsequent blocks will also have different hashes. This makes altered blocks easily detected. (ACT-IAC, 2017; Yaga et al., 2018) Fig 4 shows a generic chain of blocks.

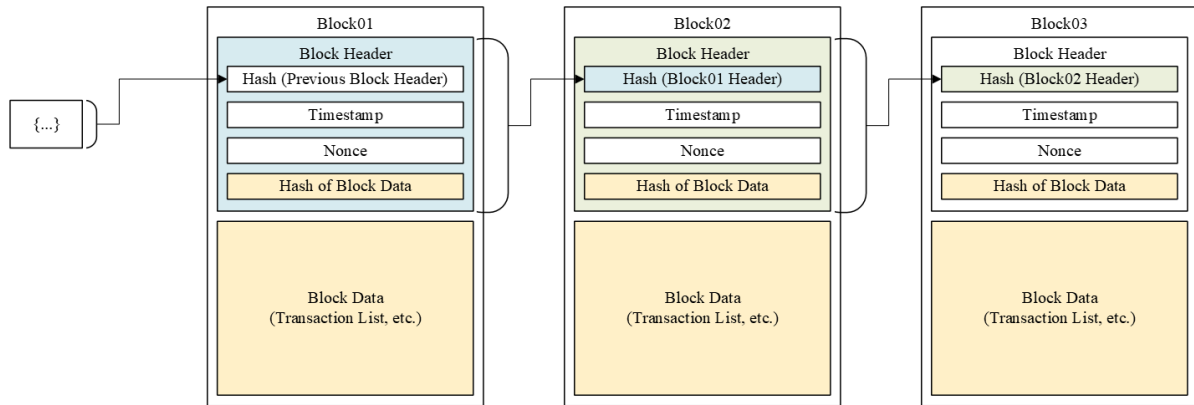


Fig. 4 Generic chain of blocks (Yaga et al., 2018)

The use of cryptographic mechanisms such as cryptographic hash functions and digital signatures ensures tamper resistant and tamper evident ledgers. Each node maintains its own copy of the ledger ensuring each node knows when a transaction has occurred. The distributed ledgers and closely linked blocks make it difficult for attackers to easily compromise the network.

3.3 Applications

Due to the immutability, transparency and security features of blockchain, BCT has been applied in various domains including the financial services, insurance sector, healthcare, voting, real estate and supply chain. (Laroiya et al., 2020)

1) **Financial services:** Barclays Bank implemented the transparency and security features of BCT in their processes (Laroiya et al., 2020).

2) **Insurance sector:** Accenture developed a blockchain system to support key processes in the insurance industry (Laroiya et al., 2020).

3) **Healthcare:** Medicalchain is the first healthcare service provider to use BCT for storing and managing automated health records, to deliver an innovative and unique telemedicine experience. MedRec uses BCT to provide secure access to patients' records (Laroiya et al., 2020).

4) **Voting:** A decentralised ledger to store voting data via BCT could prevent result manipulation by a centralised authority. Some blockchain systems in voting include BitCongress, Romotengrity and AgoraVoting (Laroiya et al., 2020).

5) **Real estate:** In the United States, Arizona and Vermont have already recognised smart contracts as a standard for real estate transactions. The BitProperty application uses the smart contract concept to allow anyone to invest in real estate, while Ubiquity implements blockchain to ensure that property data and information are securely recorded (Laroiya et al., 2020).

6) **Supply chain:** Walmart along with IBM has been working on the Hyperledger Fabric blockchain to track food staples from the suppliers to retail store shelves. Other systems include Blockverify that enforces transparency in supply chains by verifying stolen stock, counterfeit products and deceitful transactions; OriginTrain that allows consumers to trace the origin and production of products; De Beers tracks diamonds from the mine to the purchase; AidCoin that is used in nonprofit sectors; Guts that eliminates ticket fraud and secondary ticket markets by making ticket allocation transparent (Laroiya et al., 2020).

3.4 Misconceptions and Limitations of Blockchain

1 There is a tendency to overhype and overuse most emerging technologies. The following discusses some limitations and misconceptions of BCT.

1 22 1 1) *Immutability*: Many sources claim that blockchain ledgers are immutable, but blockchain is not entirely immutable and can still be modified. It is tamper evident and tamper resistant and therefore trusted. However, it can still be vulnerable to attacks, such as the 51% attack. In this attack, an attacker needs to control more than 51% of the network's resources to surpass the block creation rate of the remaining network participants. While it is possible to carry out this attack on a blockchain network, it is expensive.

28 2) *Oracle problem*: Blockchain networks work well with data within their own digital systems. (Yaga et al., 2018) However, external data such as stock prices and weather cannot be incorporated with transaction data, as it would be considered untrusted. To provide external data, blockchain networks rely on a trusted third party known as an oracle. An oracle can be any entity capable of providing reliable external data to the blockchain, such as software, sensors, or humans (ACT-IAC, 2017; Caldarelli, 2020). However, there is no way to verify whether the input data accurately reflects real-world events (Yaga et al., 2018).

3) *No trust*: Although blockchain networks are designed to function without requiring trust between users or intermediaries, a significant amount of trust is still necessary. Users must trust the cryptographic technologies utilised, trust that smart contracts operate correctly and without bugs, trust that the users of the network are not secretly colluding and trust that nodes are processing transactions fairly (Yaga et al., 2018)..

30 4) *Resource usage*: Blockchain networks that utilise the Proof of Work consensus model require significant resource and thus power consuming. (Yaga et al., 2018; Laroiya et al., 2020) Additionally, when a new full node is created, it must download a large portion or all of the blockchain data, which consumes a lot of network bandwidth.

3.5 The Enterprise System

16 An enterprise system is a vital support system that contains an organisation's data. It can include various modules such as supply chain management, manufacturing, human resource management, accounting and finance, sales, marketing and customer relationship management (Haddara et al., 2021).

Enterprise systems are centralised, while BCT is decentralised. Despite the conflict, the decentralised, immutable, irreversible and near-real-time capabilities are valuable in enterprise system processes. Therefore, blockchain can be used to enhance and reinforce enterprise systems processes. (Haddara et al., 2021)

3.6 The Supply Chain

Supply chain is the process of goods formations, from the raw materials to intermediate goods, and from final products to customers. It is closely tied to supply chain management, which involves managing materials, money, people, and information within and across the supply chain (Shukla et al., 2011). The main goal of introducing blockchain into the system is to prevent counterfeiting or theft.

36 In the context of blockchain, the goods and related documentation are the items being transacted. A digital "token" is created for physical items upon their creation. The final recipient of the item can then verify the token and trace its origin (Francisco & Swanson, 2018).

4. RECOMMENDATIONS

8 Blockchain has gained significant attention, largely due to the rise of Bitcoin that uses BCT, and the overemphasis on its immutable and irreversible properties. Many projects and organisations try to adopt the technology, even when it may not be needed (Yaga et al., 2018).

Several guidelines have been developed by agencies and organisations, including the United States Department of Homeland Security (DHS) Science & Technology Directorate, the American Council for Technology and Industry Advisory Council (ACT-IAC) and numerous white papers and articles titled "Do you need a blockchain" (Yaga et al., 2018; Wust & Gervais, 2018; Peck, 2023).

BCT is suitable for a use case that:

- Requires a shared, consistent data store (Yaga et al., 2018; Peck, 2023)
- Needs more than one entity to contribute data (Yaga et al., 2018; Wust & Gervais, 2018; Peck, 2023)
- Never update or delete data records (Yaga et al., 2018)
- Involves control issues over who runs the data store (Yaga et al., 2018; Peck, 2023)
- Requires a tamper proof log of all writes to the data store (Yaga et al., 2018; Wust & Gervais, 2018)

With this, BCT appears to be an ideal fit for supply chain management. Each stage of the supply chain involves different parties and each party may not always be trusted. A reliable and immutable record keeping system is required. Blockchain can provide an irrefutable record of every transaction with each transaction not being updated and deleted. Each party in the process can keep a copy of the record.

Despite its benefits, there are many challenges in implementing blockchain in the supply chain within enterprise. Many components would need to be integrated with existing enterprise systems. In addition, handling large amounts of data storage is required. Over time, as the number of transactions increases, the number of blocks grows, causing higher data storage demands (Solanke, n.d.; Kunduru, 2023).

CONCLUSION

Blockchain is a decentralised, peer-to-peer system that has a continuously expanding distributed ledger composed of blocks connected together by cryptographically signed transactions. Transactions are created, cryptographically signed and submitted to the blockchain network. These transactions are propagated and verified throughout the network before entering a queue to be published. A publishing node competes to add a block through a consensus model. Once the block is published, it is propagated across the network and nodes append it to their respective ledgers.

The append only ledger and the linked hash between blocks ensures that records are hardly mutable. The use of cryptography technology enhances the security of the blockchain. Thus, blockchain can foster transparency and trust in information systems.

The attention garnered by Bitcoin that uses BCT has contributed to the hype to blockchain, causing more studies and exploration of its applications. More and more enterprises and organisations are considering integrating blockchain into their processes. Blockchain has notable performance in supply chain management to trace items across the supply chain that involve many entities that may not always be trusted.

REFERENCES

- ACT-IAC. (2017). Enabling Blockchain Innovation in the U.S. Federal Government. In *ACT-IAC Emerging Technology COI*. American Council for Technology and Industry Advisory Council (ACT-IAC). https://www.actiac.org/system/files/ACT-IAC%20ENABLING%20BLOCKCHAIN%20INNOVATION_3.pdf
- Caldarelli, G. (2020). *Real-world blockchain applications under the lens of the oracle problem. A systematic literature review* (pp. 1–6). 2020 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD). <https://doi.org/10.1109/ictmod49425.2020.9380598>
- Francisco, K., & Swanson, D. (2018). The supply chain has no clothes: Technology Adoption of blockchain for supply chain transparency. *Logistics*, 2(1), 2. <https://doi.org/10.3390/logistics2010002>
- Haddara, M., Norveel, J., & Langseth, M. (2021). Enterprise systems and blockchain Technology: the dormant Potentials. *Procedia Computer Science*, 181, 562–571. <https://doi.org/10.1016/j.procs.2021.01.203>
- Hamida, E. B., Brousmiche, K. L., Levard, H., & Thea, E. (2017, July 23). *Blockchain for Enterprise: Overview, opportunities and challenges*. <https://hal.science/hal-01591859/>
- Hariyani, D., Hariyani, P., Mishra, S., & Sharma, M. K. (2025). A literature review on transformative impacts of blockchain technology on manufacturing management and industrial engineering practices. *Green Technologies and Sustainability*, 3(3), 100169. <https://doi.org/10.1016/j.grets.2025.100169>
- Hayes, A. (2025, March 24). Blockchain Facts: What Is It, How It Works, and How It Can Be Used. Investopedia. <https://www.investopedia.com/terms/b/blockchain.asp>
- Jiang, L. (2024). The use of blockchain technology in enterprise financial accounting information sharing. *PLOS ONE*, 19(2), e0298210–e0298210. <https://doi.org/10.1371/journal.pone.0298210>
- Wust, K., & Gervais, A. (2018). *Do you Need a Blockchain?* (pp. 45–54). 2018 Crypto Valley Conference on Blockchain Technology (CVCBT). <https://doi.org/10.1109/cvcbt.2018.00011>
- Kunduru, A. R. (2023). Blockchain Technology for ERP Systems: A review. *AMERICAN Journal of Engineering, Mechanics and Architecture*, 01(07), e2993-2637.

<https://khaneprozhe.ir/wp-content/uploads/2024/04/BLOCKCHAIN-TECHNOLOGY-FOR-ERP-SYS-TEMS-A-REVIEW.pdf>

- Laroiya, C., Saxena, D., & Komalavalli, C. (2020). Applications of blockchain technology. In *Elsevier eBooks* (pp. 213–243). <https://doi.org/10.1016/b978-0-12-819816-2.00009-5>
- Murtaza, A. (2023, January 18). *What is Blockchain Technology and How Does it Work?* Creative Tim's Blog - Fully Coded Design Resources for Web Developers. <https://www.creative-tim.com/blog/educational-tech/what-is-blockchain-technology-and-how-does-it-work/>
- Peck, M. E. (2023, September 20). Do you need a blockchain? *IEEE Spectrum*. <https://spectrum.ieee.org/do-you-need-a-blockchain>
- Shukla, R. K., Garg, D., & Agarwal, A. (2011). Understanding of Supply Chain: A Literature Review. *International Journal of Engineering Science and Technology (IJEST)*, 3(3), 2059–2072. https://www.researchgate.net/publication/50946316_UNDERSTANDING_OF_SUPPLY_CHAIN_A_LITERATURE_REVIEW
- Solanke, A. (n.d.). *Blockchain implementation for enterprise data integrity: distributed ledger strategies for supply chain transparency and trustworthy data verification* [Dallas Baptist University]. https://www.researchgate.net/profile/Adedamola-Solanke/publication/390212745_Blockchain_Implementation_for_Enterprise_Data_Integrity_Distributed_Ledger_Strategies_for_Supply_Chain_Transparency_and_Trustworthy_Data_Verification/links/67e49df58a5ab03f971310b5/Blockchain-Implementation-for-Enterprise-Data-Integrity-Distributed-Ledger-Strategies-for-Supply-Chain-Transparency-and-Trustworthy-Data-Verification.pdf
- Popovski, L., Soussou, G., & Webb, P. B. (2018). A brief history of blockchain. *Legaltech news*, 3.
- Wu, H., Cao, J., Yang, Y., Tung, C. L., Jiang, S., Tang, B., Liu, Y., Wang, X., & Deng, Y. (2019). Data Management in Supply Chain Using Blockchain: Challenges and a Case Study. 2019 28th International Conference on Computer Communication and Networks (ICCCN). <https://doi.org/10.1109/icccn.2019.8846964>
- Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2018). Blockchain technology overview. In *National Institute of Standards and Technology Internal Report 8202*. <https://doi.org/10.6028/nist.ir.8202>

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Blockchain for Supply Chain Transparency and Trust in Enterprise Systems

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Abstract — *This paper presents the integration of blockchain technology into enterprise systems, specifically supply chain to enhance data transparency and trust. It investigates the background, workings and real-world applications of blockchain technology. The study concludes that blockchain has a great potential to support supply chains in enterprise systems.*

Keywords— *Blockchain, Supply Chain, Enterprise System, Transparency, Security*

EXECUTIVE SUMMARY

This paper explores the integration of blockchain technology into enterprise systems to maintain the transparency and security of the information. The first part of the report introduces what blockchain is, continues with investigates its background and discusses challenges while implementing it into current enterprise systems. Blockchain, initially developed to support Bitcoin as a decentralized platform to store all the data securely and keep it synchronized. Until now, its application has then extended to various industries other than the financial field, such as healthcare, supply chain management and logistics. Although blockchain offers many benefits, it is still not widely adopted by most companies, mainly due to the lack of technical expertise and infrastructure support.

The second part of the report explains blockchain's core mechanisms and examines how it improves the transparency and trust for the enterprise performance. Furthermore, the report also reviews the real-world application across various domains, discussing the advantages and disadvantages of blockchain adoption and lastly provides some recommendations for the enterprise. In conclusion, blockchain technology has shown great potential to improve transparency, and trust for supply chains in enterprise systems.

1. INTRODUCTION

In 2009, blockchain first gained popularity with the introduction of Bitcoin in the market and was originally designed to prevent double spending in digital currencies. It began as a secure, transparent, and decentralized platform for peer-to-peer transactions and the recording of information, also known as the foundation of cryptocurrency (Popovski, Soussou, & Webb, 2018). This technology has grown rapidly and become more crucial in enterprise systems, particularly in supply chain management because transparency and trust are important elements for effective collaboration between stakeholders. As an example, tracking goods, verifying transactions are covered by blockchain to ensure accurate and reliable information shared across organizations, especially for multi-partner organizations.

As technology evolves, blockchain becomes a great idea to enhance enterprise operational capabilities. In a recent quantitative investigation, blockchain sharing methods are being proven that perform better than traditional ways in every aspect, including information-sharing efficiency, data accuracy and cost reduction (Jiang, 2024). These improvements can significantly reduce operational errors, allow faster decision making, better resource monitoring and more reliable reporting. Besides to boost the organizational performance and facilitate the business operations, blockchain technology can also lead to transparency and trust improvement within enterprises. In short, blockchain technology acts as a strong foundation for companies systems by keeping their data secure and supporting automated processes.

However, the use of the blockchain in enterprise systems remains limited with several types of issues, such as organizational, ethical, and technical difficulties. In order to better understand blockchain technology, this article will primarily examine its architecture, use cases, benefits, limitations and potential future development. It also aims to offer insight to researchers and organizations who are interested about how blockchain is being used in enterprise settings.

2. BACKGROUND

A blockchain defined as a distributed database shared across a computer nodes network. It is best known in cryptocurrency systems such as Bitcoin, where it enables a safe and decentralized record of transactions (Hayes, 2025). However, blockchain technology is definitely not only limited to cryptocurrency, it also involves finance, healthcare, supply chain management, etc. These characteristics explain why it has gained popularity by producing transparent, tamper-proof, and verifiable transaction records without relying on centralized authorities.

Blockchain technology (BCT) is currently being considered a transformative innovation to increase transparency, traceability, and trust in enterprise systems, particularly supply chain management. Traditional supply chains before this relied on centralized databases, which are often vulnerable to fraud, forgery, and delays (Wu et al., 2019) because the stakeholders like suppliers, manufacturers, distributors, and retailers have restricted exposure into accurate, secure information. However, the decentralized architecture of BCT that securely and consistently records every transaction ensures the same truth for them. (Hariyani et al., 2025).

The studies show that BCT may solve inefficiencies in traditional supply chain systems. For example, in Francisco and Swanson (2018), the researchers study blockchain adoption and evaluate user acceptance using the Unified Theory of Acceptance and Use of Technology (UTAUT) framework based on performance expectancy, effort expectancy, social influence, and facilitating conditions. According to the findings, many organizations are still unwilling to implement BCT due to uncertain return on investment, unclear regulatory from administrative and most important, the lack of technical expertise to integrate it with legacy systems.

Additionally, the researchers also examined how BCT interacts with other technologies to enhance enterprise performance. In the era of Industry 4.0, BCT combines with artificial intelligence (AI) and the Internet of Things (IoT) to provide secure and immutable data for predictive maintenance and real-time inventory management (Hariyani et al., 2025). Besides, smart contracts are used to integrate BCT with automation to support the Circular Economy and sustainable energy trade. As a result, an organization that integrates BCT with other digital technologies becomes more flexible and efficient.

In summary, BCT offers a better solution to solve the problems faced by traditional supply chains. Although its integration with emerging technologies such as AI and IoT enhances enterprise performance and data integrity, the adoption issues still remain. Therefore, future research should focus on developing standardized frameworks and practical integration strategies to fully utilize blockchain.

3. ANALYSIS AND DISCUSSION

3.1 The Blockchain Technology

Blockchain is a trustless, peer-to-peer and ever-expanding ledger of records, including smart contracts, which are executed and shared among involved parties (Hamida et al., 2017). Blockchain, as its name suggests, is a chain of blocks that are cryptographically linked forming a ledger, with each block containing a group of cryptographically signed transactions (ACT-IAC, 2017; Yaga et al., 2018) as shown in Fig. 1.

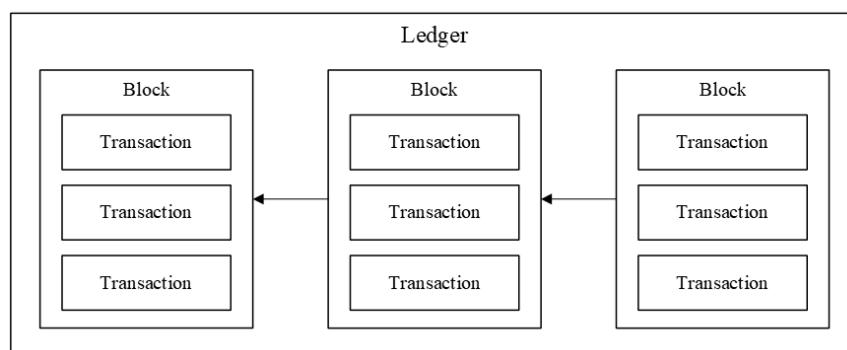


Fig. 1 Transactions are grouped into blocks and referred to as the ledger.

Blockchain relies on peer-to-peer networks. The entire blockchain network is made of peers or nodes. The ledger is distributed to each node in the blockchain network (Yaga et al., 2018) as shown in Fig 2.

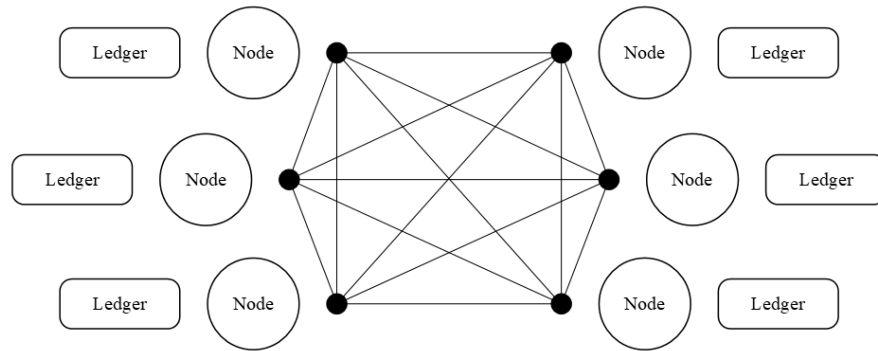


Fig. 2 A blockchain network made up of nodes, with each node holding a copy of the ledger (Murtaza, 2023).

A blockchain network user submits candidate transactions using softwares such as applications, digital wallets and web services. These softwares send the transactions to one or more nodes within the blockchain network. The submitted transactions are then propagated to other nodes across the network. Once it reaches a publishing node, it must wait in a queue until the publishing node adds it to a block and publishes the block to the blockchain. Multiple publishing nodes may compete to publish the next block at the same time and a consensus model such as Proof Of Work and Proof of Stake is used to determine which node publishes the next block. The transaction is validated by ensuring it is correctly formatted and cryptographically signed by the provider of the digital asset. Invalid transactions are ignored, preventing them from propagating throughout the network. Other nodes will check the validity of the transactions in the published block before adding the block to its own copy of the ledger (Yaga et al., 2018).

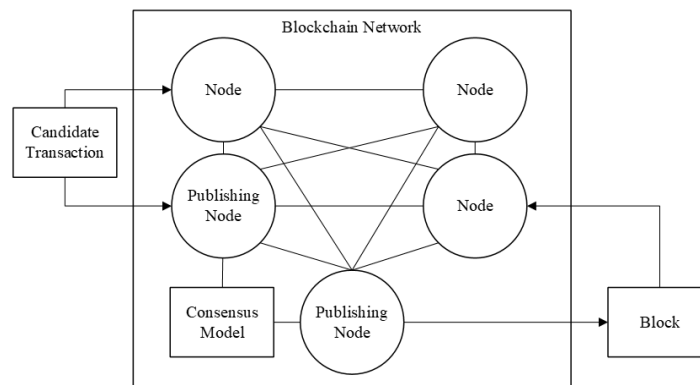


Fig. 3 A transaction is added to a block

3.2 How Blockchain is Transparent and Reliable

The architecture of blockchain is designed for use in environments where participants may not fully trust one another. The security measures and transparent record keeping are the advantages of blockchain technology. The following components help ensure that blockchain is secure and reliable.

1) *Cryptographic hash functions*: Blockchain technology utilises cryptographic hash functions for many operations. Input data is processed to generate a unique output or digest. Cryptographic hash are preimage resistant, where the input cannot be computed given the digest; and collision resistant, where one cannot find two inputs that hash to the same output (Yaga et al., 2018).

2) *Asymmetric-key cryptography*: This method uses a mathematically related pair of public and private keys. Despite the relationship, the private key cannot be efficiently derived from the public key. Each transaction must be 'digitally signed', where the private key encrypts the transaction and anyone with the public key can decrypt it and thus, verifying the signature generated by the private key. Encrypting the transaction with the private key proves that the signer has access to the private key. (Yaga et al., 2018)

3) *Ledgers*: A ledger is a collection of transactions and is distributed among the network, with each user maintaining their own copy. When new nodes join the blockchain network, they discover other full nodes and request a full copy of the ledger. This creates many backup copies and all nodes updating and syncing the same ledger data. Due to its distributed nature, the blockchain network is heterogeneous, where the hardware,

software and network infrastructure vary between nodes. An attack on a node may not work on other nodes. Additionally, there is no centralised point of attack. (Yaga et al., 2018)

4) *Blocks*: A block consists of a block header and block data. The block header stores metadata for the block and the block data contains the list of validated candidate transactions or other data. Blocks are chained together and each block contains the hash digest of the previous block's header. If a previously published block is changed, it will generate a different hash and all subsequent blocks will also have different hashes. This makes altered blocks easily detected. (ACT-IAC, 2017; Yaga et al., 2018) Fig 4 shows a generic chain of blocks.

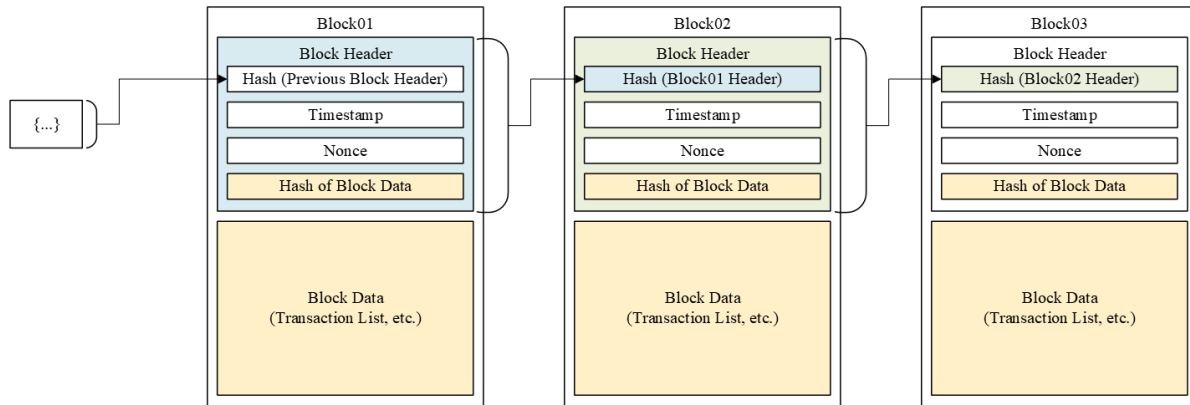


Fig. 4 Generic chain of blocks (Yaga et al., 2018)

The use of cryptographic mechanisms such as cryptographic hash functions and digital signatures ensures tamper resistant and tamper evident ledgers. Each node maintains its own copy of the ledger ensuring each node knows when a transaction has occurred. The distributed ledgers and closely linked blocks make it difficult for attackers to easily compromise the network.

3.3 Applications

Due to the immutability, transparency and security features of blockchain, BCT has been applied in various domains including the financial services, insurance sector, healthcare, voting, real estate and supply chain. (Laroiya et al., 2020)

1) *Financial services*: Barclays Bank implemented the transparency and security features of BCT in their processes (Laroiya et al., 2020).

2) *Insurance sector*: Accenture developed a blockchain system to support key processes in the insurance industry (Laroiya et al., 2020).

3) *Healthcare*: Medicalchain is the first healthcare service provider to use BCT for storing and managing automated health records, to deliver an innovative and unique telemedicine experience. MedRec uses BCT to provide secure access to patients' records (Laroiya et al., 2020).

4) *Voting*: A decentralised ledger to store voting data via BCT could prevent result manipulation by a centralised authority. Some blockchain systems in voting include BitCongress, Romotengrity and AgoraVoting (Laroiya et al., 2020).

5) *Real estate*: In the United States, Arizona and Vermont have already recognised smart contracts as a standard for real estate transactions. The BitProperty application uses the smart contract concept to allow anyone to invest in real estate, while Ubiquity implements blockchain to ensure that property data and information are securely recorded (Laroiya et al., 2020).

6) *Supply chain*: Walmart along with IBM has been working on the Hyperledger Fabric blockchain to track food staples from the suppliers to retail store shelves. Other systems include Blockverify that enforces transparency in supply chains by verifying stolen stock, counterfeit products and deceitful transactions; OriginTrain that allows consumers to trace the origin and production of products; De Beers tracks diamonds from the mine to the purchase; AidCoin that is used in nonprofit sectors; Guts that eliminates ticket fraud and secondary ticket markets by making ticket allocation transparent (Laroiya et al., 2020).

3.4 Misconceptions and Limitations of Blockchain

There is a tendency to overhype and overuse most emerging technologies. The following discusses some limitations and misconceptions of BCT.

1) *Immutability*: Many sources claim that blockchain ledgers are immutable, but blockchain is not entirely immutable and can still be modified. It is tamper evident and tamper resistant and therefore trusted. However, it can still be vulnerable to attacks, such as the 51% attack. In this attack, an attacker needs to control more than 51% of the network's resources to surpass the block creation rate of the remaining network participants. While it is possible to carry out this attack on a blockchain network, it is expensive.

2) *Oracle problem*: Blockchain networks work well with data within their own digital systems. (Yaga et al., 2018) However, external data such as stock prices and weather cannot be incorporated with transaction data, as it would be considered untrusted. To provide external data, blockchain networks rely on a trusted third party known as an oracle. An oracle can be any entity capable of providing reliable external data to the blockchain, such as software, sensors, or humans (ACT-IAC, 2017; Caldarelli, 2020). However, there is no way to verify whether the input data accurately reflects real-world events (Yaga et al., 2018).

3) *No trust*: Although blockchain networks are designed to function without requiring trust between users or intermediaries, a significant amount of trust is still necessary. Users must trust the cryptographic technologies utilised, trust that smart contracts operate correctly and without bugs, trust that the users of the network are not secretly colluding and trust that nodes are processing transactions fairly (Yaga et al., 2018)..

4) *Resource usage*: Blockchain networks that utilise the Proof of Work consensus model require significant resource and thus power consuming. (Yaga et al., 2018; Laroiya et al., 2020) Additionally, when a new full node is created, it must download a large portion or all of the blockchain data, which consumes a lot of network bandwidth.

3.5 The Enterprise System

An enterprise system is a vital support system that contains an organisation's data. It can include various modules such as supply chain management, manufacturing, human resource management, accounting and finance, sales, marketing and customer relationship management (Haddara et al., 2021).

Enterprise systems are centralised, while BCT is decentralised. Despite the conflict, the decentralised, immutable, irreversible and near-real-time capabilities are valuable in enterprise system processes. Therefore, blockchain can be used to enhance and reinforce enterprise systems processes. (Haddara et al., 2021)

3.6 The Supply Chain

Supply chain is the process of goods formations, from the raw materials to intermediate goods, and from final products to customers. It is closely tied to supply chain management, which involves managing materials, money, people, and information within and across the supply chain (Shukla et al., 2011). The main goal of introducing blockchain into the system is to prevent counterfeiting or theft.

In the context of blockchain, the goods and related documentation are the items being transacted. A digital "token" is created for physical items upon their creation. The final recipient of the item can then verify the token and trace its origin (Francisco & Swanson, 2018).

4. RECOMMENDATIONS

Blockchain has gained significant attention, largely due to the rise of Bitcoin that uses BCT, and the overemphasis on its immutable and irreversible properties. Many projects and organisations try to adopt the technology, even when it may not be needed (Yaga et al., 2018).

Several guidelines have been developed by agencies and organisations, including the United States Department of Homeland Security (DHS) Science & Technology Directorate, the American Council for Technology and Industry Advisory Council (ACT-IAC) and numerous white papers and articles titled "Do you need a blockchain" (Yaga et al., 2018; Wust & Gervais, 2018; Peck, 2023).

BCT is suitable for a use case that:

- Requires a shared, consistent data store (Yaga et al., 2018; Peck, 2023)
- Needs more than one entity to contribute data (Yaga et al., 2018; Wust & Gervais, 2018; Peck, 2023)
- Never update or delete data records (Yaga et al., 2018)
- Involves control issues over who runs the data store (Yaga et al., 2018; Peck, 2023)
- Requires a tamper proof log of all writes to the data store (Yaga et al., 2018; Wust & Gervais, 2018)

With this, BCT appears to be an ideal fit for supply chain management. Each stage of the supply chain involves different parties and each party may not always be trusted. A reliable and immutable record keeping system is required. Blockchain can provide an irrefutable record of every transaction with each transaction not being updated and deleted. Each party in the process can keep a copy of the record.

Despite its benefits, there are many challenges in implementing blockchain in the supply chain within enterprise. Many components would need to be integrated with existing enterprise systems. In addition, handling large amounts of data storage is required. Over time, as the number of transactions increases, the number of blocks grows, causing higher data storage demands (Solanke, n.d.; Kunduru, 2023).

CONCLUSION

Blockchain is a decentralised, peer-to-peer system that has a continuously expanding distributed ledger composed of blocks connected together by cryptographically signed transactions. Transactions are created, cryptographically signed and submitted to the blockchain network. These transactions are propagated and verified throughout the network before entering a queue to be published. A publishing node competes to add a block through a consensus model. Once the block is published, it is propagated across the network and nodes append it to their respective ledgers.

The append only ledger and the linked hash between blocks ensures that records are hardly mutable. The use of cryptography technology enhances the security of the blockchain. Thus, blockchain can foster transparency and trust in information systems.

The attention garnered by Bitcoin that uses BCT has contributed to the hype to blockchain, causing more studies and exploration of its applications. More and more enterprises and organisations are considering integrating blockchain into their processes. Blockchain has notable performance in supply chain management to trace items across the supply chain that involve many entities that may not always be trusted.

REFERENCES

- ACT-IAC. (2017). Enabling Blockchain Innovation in the U.S. Federal Government. In *ACT-IAC Emerging Technology COI*. American Council for Technology and Industry Advisory Council (ACT-IAC). https://www.actiac.org/system/files/ACT-IAC%20ENABLING%20BLOCKCHAIN%20INNOVATION_3.pdf
- Caldarelli, G. (2020). *Real-world blockchain applications under the lens of the oracle problem. A systematic literature review* (pp. 1–6). 2020 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD). <https://doi.org/10.1109/ictmod49425.2020.9380598>
- Francisco, K., & Swanson, D. (2018). The supply chain has no clothes: Technology Adoption of blockchain for supply chain transparency. *Logistics*, 2(1), 2. <https://doi.org/10.3390/logistics2010002>
- Haddara, M., Norveel, J., & Langseth, M. (2021). Enterprise systems and blockchain Technology: the dormant Potentials. *Procedia Computer Science*, 181, 562–571. <https://doi.org/10.1016/j.procs.2021.01.203>
- Hamida, E. B., Brousmiche, K. L., Levard, H., & Thea, E. (2017, July 23). *Blockchain for Enterprise: Overview, opportunities and challenges*. <https://hal.science/hal-01591859/>
- Hariyani, D., Hariyani, P., Mishra, S., & Sharma, M. K. (2025). A literature review on transformative impacts of blockchain technology on manufacturing management and industrial engineering practices. *Green Technologies and Sustainability*, 3(3), 100169. <https://doi.org/10.1016/j.grets.2025.100169>
- Hayes, A. (2025, March 24). Blockchain Facts: What Is It, How It Works, and How It Can Be Used. Investopedia. <https://www.investopedia.com/terms/b/blockchain.asp>
- Jiang, L. (2024). The use of blockchain technology in enterprise financial accounting information sharing. *PLOS ONE*, 19(2), e0298210–e0298210. <https://doi.org/10.1371/journal.pone.0298210>
- Wust, K., & Gervais, A. (2018). *Do you Need a Blockchain?* (pp. 45–54). 2018 Crypto Valley Conference on Blockchain Technology (CVCBT). <https://doi.org/10.1109/cvcbt.2018.00011>
- Kunduru, A. R. (2023). Blockchain Technology for ERP Systems: A review. *AMERICAN Journal of Engineering, Mechanics and Architecture*, 01(07), e2993-2637.

<https://khaneprozhe.ir/wp-content/uploads/2024/04/BLOCKCHAIN-TECHNOLOGY-FOR-ERP-SYS-TEMS-A-REVIEW.pdf>

- Laroiya, C., Saxena, D., & Komalavalli, C. (2020). Applications of blockchain technology. In *Elsevier eBooks* (pp. 213–243). <https://doi.org/10.1016/b978-0-12-819816-2.00009-5>
- Murtaza, A. (2023, January 18). *What is Blockchain Technology and How Does it Work?* Creative Tim's Blog - Fully Coded Design Resources for Web Developers. <https://www.creative-tim.com/blog/educational-tech/what-is-blockchain-technology-and-how-does-it-work/>
- Peck, M. E. (2023, September 20). Do you need a blockchain? *IEEE Spectrum*. <https://spectrum.ieee.org/do-you-need-a-blockchain>
- Shukla, R. K., Garg, D., & Agarwal, A. (2011). Understanding of Supply Chain: A Literature Review. *International Journal of Engineering Science and Technology (IJEST)*, 3(3), 2059–2072. https://www.researchgate.net/publication/50946316_UNDERSTANDING_OF_SUPPLY_CHAIN_A_LITERATURE_REVIEW
- Solanke, A. (n.d.). *Blockchain implementation for enterprise data integrity: distributed ledger strategies for supply chain transparency and trustworthy data verification* [Dallas Baptist University]. https://www.researchgate.net/profile/Adedamola-Solanke/publication/390212745_Blockchain_Implementation_for_Enterprise_Data_Integrity_Distributed_Ledger_Strategies_for_Supply_Chain_Transparency_and_Trustworthy_Data_Verification/links/67e49df58a5ab03f971310b5/Blockchain-Implementation-for-Enterprise-Data-Integrity-Distributed-Ledger-Strategies-for-Supply-Chain-Transparency-and-Trustworthy-Data-Verification.pdf
- Popovski, L., Soussou, G., & Webb, P. B. (2018). A brief history of blockchain. *Legaltech news*, 3.
- Wu, H., Cao, J., Yang, Y., Tung, C. L., Jiang, S., Tang, B., Liu, Y., Wang, X., & Deng, Y. (2019). Data Management in Supply Chain Using Blockchain: Challenges and a Case Study. 2019 28th International Conference on Computer Communication and Networks (ICCCN). <https://doi.org/10.1109/icccn.2019.8846964>
- Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2018). Blockchain technology overview. In *National Institute of Standards and Technology Internal Report 8202*. <https://doi.org/10.6028/nist.ir.8202>